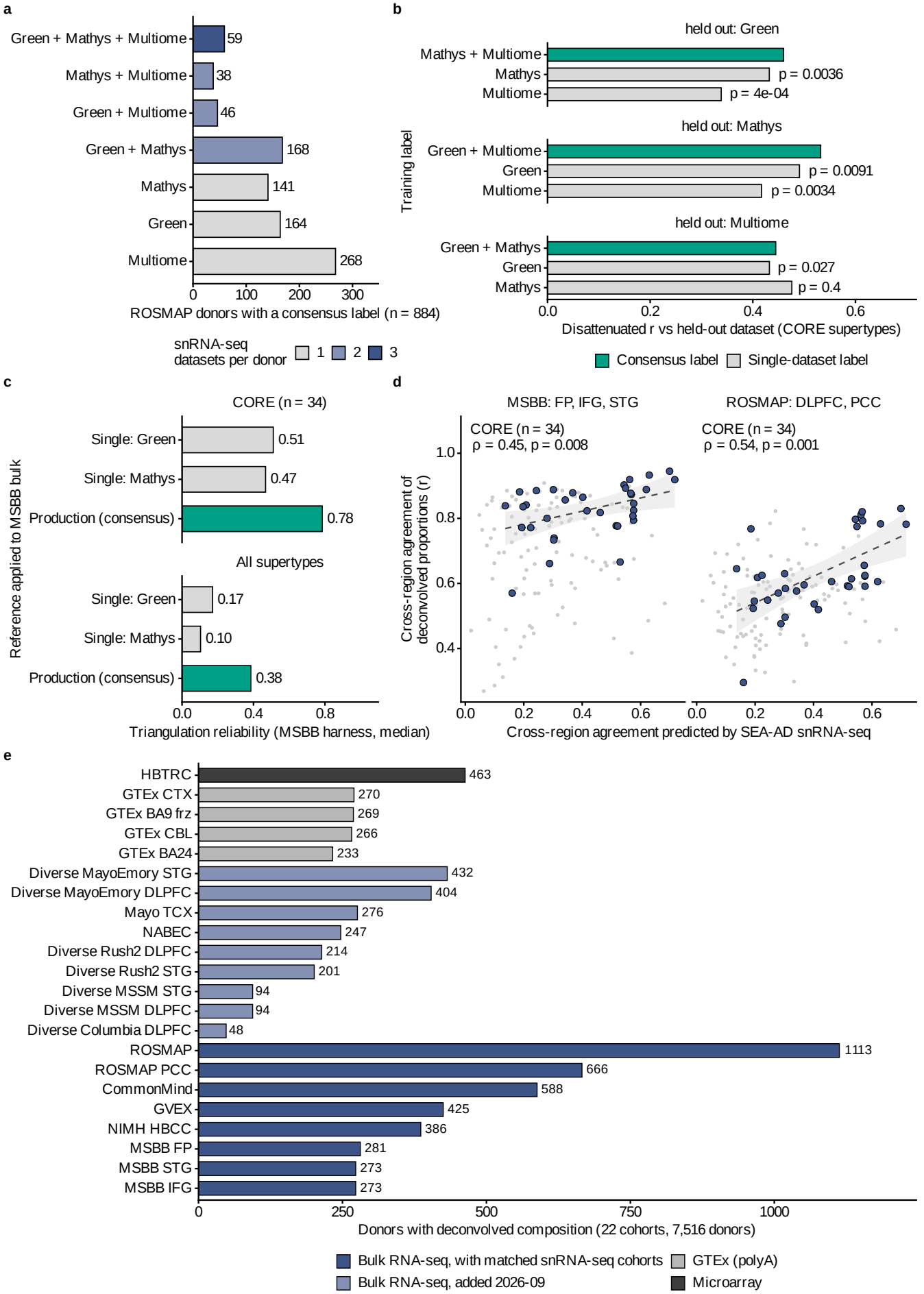
a
Green + Mathys + Multiome
59
Mathys + Multiome
38
Green + Multiome
46
Green + Mathys
168
Mathys
141
Green
164
Multiome
268
0
100
200
300
ROSMAP donors with a consensus label (n = 884)
snRNA-seq datasets per donor
1
2
3
b
held out: Green
Mathys + Multiome
Mathys
p = 0.0036
Multiome
p = 4e-04
held out: Mathys
Green + Multiome
Green
p = 0.0091
Multiome
p = 0.0034
held out: Multiome
Green + Mathys
Green
p = 0.027
Mathys
p = 0.4
Training label
0.0
0.2
0.4
0.6
Disattenuated r vs held-out dataset (CORE supertypes)
Consensus label
Single-dataset label
c
CORE (n = 34)
Single: Green
0.51
Single: Mathys
0.47
Production (consensus)
0.78
All supertypes
Single: Green
0.17
Single: Mathys
0.10
Production (consensus)
0.38
Reference applied to MSBB bulk
0.0
0.4
0.8
Triangulation reliability (MSBB harness, median)
d
MSBB: FP, IFG, STG
ROSMAP: DLPFC, PCC
CORE (n = 34)
ρ = 0.45, p = 0.008
CORE (n = 34)
ρ = 0.54, p = 0.001
Cross-region agreement of deconvolved proportions (r)
1.0
0.8
0.6
0.4
0.0
0.2
0.4
0.6
Cross-region agreement predicted by SEA-AD snRNA-seq
e
HBTRC
463
GTEx CTX
270
GTEx BA9 frz
269
GTEx CBL
266
GTEx BA24
233
Diverse MayoEmory STG
432
Diverse MayoEmory DLPFC
404
Mayo TCX
276
NABEC
247
Diverse Rush2 DLPFC
214
Diverse Rush2 STG
201
Diverse MSSM STG
94
Diverse MSSM DLPFC
94
Diverse Columbia DLPFC
48
ROSMAP
1113
ROSMAP PCC
666
CommonMind
588
GVEX
425
NIMH HBCC
386
MSBB FP
281
MSBB STG
273
MSBB IFG
273
0
250
500
750
1000
Donors with deconvolved composition (22 cohorts, 7,516 donors)
Bulk RNA-seq, with matched snRNA-seq cohorts
GTEx (polyA)
Bulk RNA-seq, added 2026-09
Microarray